



The standard dynamic energy budget model has no plausible alternatives

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ABSTRACT

During its 40 years of development, the assumptions of the Dynamic Energy Budget (DEB) theory for metabolic organisation turned out hard to replace. To understand this, a reasoning is here presented for why its standard model has no alternatives with a comparable level of simplicity and will never have them. Energy and mass conservation rules are essential to quantify the eco-physiological development of an individual organism thermodynamically. These rules strongly constrain the mathematical modelling of this development. In combination with consistency with a small set of stylised empirical facts, the freedom of modelling the skeleton of the model is reduced to a single one: the standard DEB model. This skeleton can, however, be extended in many different ways to capture particular ‘details’. The key-message of this paper is that the more simple metabolic models become, the more constraining are consistency conditions.

1. Introduction

Dynamic Energy Budget (DEB) theory aims to specify the metabolism of an individual thermodynamically in a changing environment, in terms of temperature and substrate availability, where metabolism includes feeding, digestion, maintenance, development, growth, reproduction, respiration, excretion and, possibly, aging.

The theory has been set-up as a formal theory, where a set of consistent and coherent assumptions specify mathematical models. Many models for various aspects of metabolism have been proposed, but the key issue revealed by DEB theory is that all these aspects have to be modelled effectively at the same time to specify the individual (thermo) dynamically. These processes do interact and have to be studied in coherence, while being consistent and realistic as well as general, i.e. not species-specific. This problem is much more complex, compared to modelling a single aspect of energetics only, and restricts the possible classes of models substantially.

Originally, in 1979, I intended to set up a DEB model based on a set of assumptions, as listed in Table 1, and subsequently replace assumptions to arrive at a set of different models that can be compared for realism and other criteria. This is typically an effective way to gain insight. Despite considerable effort over the years, it turned out to be very hard to replace assumptions, however, which motivated research for the reasons. The standard DEB model can be considered to be a canonical form among DEB models, and thought to be applicable to many animal species, which are metabolically simpler than other taxa due to the fact that they live off complex substrates, namely other organisms. Introductions to the standard DEB model can be found in

Jusup et al. (2017); Kooijman (2001, 2012); Ledder (2014); van der Meer (2006); Muller et al. (2019); Sara et al. (2014); Sousa et al. (2008, 2010), an evolutionary context in Kooijman et al. (2003); Kooijman and Troost (2007) and a system-earth context in Kooijman (2004). Other DEB models are extensions of the standard one, including multiple substrates, reserves and structures, extra life stages, foetal development, etc.

Although no new elements in the development of DEB theory showed up in the context of understanding why it has no alternatives, the reasoning in the following section is new, and is meant to be an open invitation to find holes.

2. Reasoning that concludes uniqueness

The following 7-step reasoning leads to the conclusion that it is plausible that alternatives for the standard DEB model are impossible with a comparable level of simplicity and consistency with a small selection of the stylised facts that are listed in Table 2.

1. To specify the individual thermodynamically, we need mass balances for chemical elements. Since life exists in anaerobic environments (Fenchel and Finlays, 1995), the use of dioxygen cannot be used as an universal quantifier for metabolic rate. Since some microbes absorb heat, rather than produce it (Liu et al., 2001), heat dissipation can also not be used as quantifier. The only quantifier that works is entropy dissipation (Sousa et al., 2006), but to access entropy of living biomass, we need an entropy balance, and, therefore, an energy balance. To create an energy balance, however,

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Table 1

The assumptions that specify the standard DEB model. Taken from Kooijman (2010, Table 2.4).

1. The amounts of reserve, structure and maturity are the state variables of the individual; reserve and structure have a constant composition (strong homeostasis) and maturity represents information.
2. Substrate (food) uptake is initiated (birth) and allocation to maturity is redirected to reproduction (puberty) if maturity reaches certain threshold values.
3. Food is converted into reserve and reserve is mobilised at a rate that depends on the state variables only to fuel all other metabolic processes.
4. The embryonic stage has initially a negligibly small amount of structure and maturity (but a substantial amount of reserve). The reserve density at birth equals that of the mother at egg formation (maternal effect). Foetuses develop in the same way as embryos in eggs, but at a rate unrestricted by reserve availability.
5. The feeding rate is proportional to the surface area of the individual and the food-handling time is independent of food density.
6. The reserve density at constant food density does not depend on the amount of structure (weak homeostasis).
7. Somatic maintenance is proportional to structural volume, but some components (osmosis in aquatic organisms, heating in endotherms) are proportional to structural surface area.
8. Maturity maintenance is proportional to the level of maturity
9. A fixed fraction of mobilised reserves is allocated to somatic maintenance plus growth, the rest to maturity maintenance plus maturation or reproduction (the κ -rule).
10. The individual does not change in shape during growth (isomorphism). This assumption applies to the standard DEB model only.

we need a mass balance, see Kooijman (2010, Section 4.8.1).

2. To create a mass balance, we need the strong homeostasis assumption: pools of metabolites that do not change in chemical composition. This excludes biochemical models that follow particular chemical compounds, because we need to follow *all* chemical compounds in an individual to create a mass balance, which is impossible. So the notion of pools of metabolites is unavoidable.
3. Given strong homeostasis, we need at least 2 pools. With a single pool, it will not be possible, for instance, to include (metabolic) memory, such as the nutritional status (c.f. Table 2 F2) or smoothing out fluctuations in food availability, and embryo development. Kooijman (2010, Section 1.1.3) gives additional arguments for why we minimally need 2 pools.
4. We need the weak homeostasis assumption, i.e. constant chemical composition of the whole body during growth in constant environments, to access the chemical composition of the pools for constructing mass balances (at least in terms of frequencies of chemical elements). Like the strong homeostasis assumption, it is unavoidable. Moreover it matches stylised fact S2 in Table 2.
5. Within the class of 2-pool models with weak homeostasis, the standard DEB model is the only one among the 21 topological alternatives that is consistent with a set of 4 stylised facts as listed in Table 2, see Lika and Kooijman (2011): F2,G1,G2,S2. These alternatives include models with varying fraction of mobilised reserve that is allocated to soma (c.f. Table 1 assumption 9).
6. This still allows freedom in the relationship between food availability and ingestion rate. The Synthesizing Units, that DEB theory uses for specification (Kooijman, 1998), represent a very natural framework in the context of Markov processes, that can readily accommodate behavioural modifications (Haccou and Meelis, 1992).
7. The final bit of freedom is how food intake relates to size. The DEB assumption that food intake is proportional to surface area has a nice link with underlying physics for transport, while the simplest, and parameter-free, way to relate surface area to volume results from the assumption of the standard DEB model that shape does not change during growth (isomorphy). This can readily be extended to include particular changes in shape during ontogeny, such as done to capture metabolic acceleration (Kooijman, 2014).

Table 2

Stylised empirical facts on energetics, modified from Sousa et al. (2008).

Feeding

- F1 Many species (almost all animals and plants) have an embryo stage that does not feed
- F2 During starvation, organisms are able to reproduce, grow and survive for some time
- F3 At abundant food, the feeding rate is at some maximum, independent of food density

Growth

- G1 Length of isomorphic organisms at abundant food is a (fully) satiating function of age; at constant food density, no substantial shrinking occurs independent of ageing
- G2 The inverse generalised von Bertalanffy growth rate (see Appendix) increases linearly with ultimate length both intra-specifically (or different constant food levels) and inter-specifically for large body sizes
- G3 Foetuses increase in weight approximately proportional to cubed time

Reproduction

- R1 Many species (almost all animals and plants) have a juvenile stage that does not reproduce
- R2 Reproduction increases with size intra-specifically, but decreases with size inter-specifically
- R3 A range of constant low food levels exists at which an individual can survive, but not reproduce
- R4 Growth can be simultaneous with reproduction, but growth can also cease long before reproduction is initiated

Respiration

- O1 Animal eggs and plant seeds initially hardly use dioxygen
- O2 The use of dioxygen increases with decreasing mass in embryos and increases with mass in juveniles and adults
- O3 The use of dioxygen by isomorphs scales approximately with body weight raised to a power close to 0.75
- O4 Animals show a transient increase in metabolic rate after ingesting food (heat increment of feeding)

Stoichiometry

- S1 The chemical composition of organisms depends on the nutritional status (starved vs well-fed)
- S2 The chemical composition of organisms at constant food density of constant composition becomes constant during growth

Energy

- E1 Some energy always dissipates, also in absence of dioxygen
- E2 Dissipating heat for heterotrophic isomorphs under aerobic conditions is a weighted sum of three mass flows: carbon dioxide, dioxygen and nitrogenous waste

Ageing

- A1 Mean life span typically increases inter-specifically with maximum body length in endotherms, but hardly depends on body length in ectotherms
- A2 Survivor curves for life span terminated by ageing are typically well described by the Weibull and Gompertz models

The reasoning involves only a few of the facts in Table 2, but DEB theory is found to be consistent with all of them (Sousa et al., 2008). I do realise that this reasoning partly shifts the discussion from modelling to stylised empirical facts, but I see this as a substantial contribution to structuring thoughts. It is conceivable that, with a different set of stylised facts, another, also possibly unique, most simple model would be implied. But, since fact S2 is unavoidable for the reasons given in step 4, and mild deviations from fact G2 have no consequences, while facts F2 and G1 are robust, this seems to be rather academic. The significance of stylised facts is directly linked to the unproven presumption of DEB theory that it is possible to construct a theory on energetics/metabolism that is not species-specific. Stylised facts play a key role in how to test theories for realism that are not species-specific, in combination with data from a large set of single species. Some biologists will doubtlessly argue that such theories are impossible, and biology is bound to remain a descriptive discipline. Yet it seems equally hard to demonstrate convincingly that such theories are *not* possible. My own position is that, if you do not work on this, you will never get it. All theories have elements of stylization, which involve the context in judging differences between data and predictions. But in view of the next paragraph, I am optimistic about the possible existence of such a theory, at least for a range of applications.

A remarkable observation in the context of lack of alternatives for

Table 3

Empirical models that turn out to be special cases of DEB models, or very good numerical approximations to them. Modified from Kooijman (2010, Table 11.1).

author (year)	model
Lavoisier and Laplace (1780)	multiple regression of heat against mineral fluxes
Gompertz (1825)	survival probability for ageing
Arrhenius (1889)	temperature dependence of physiological rates
Huxley (1932); Snell (1891)	allometric growth of body parts
Henri (1902)	Michaelis–Menten kinetics
Blackman (1905)	bilinear functional response
Hill (1910)	Hill's functional response
Thornton (1917)	heat dissipation
Pütter (1920)	von Bertalanffy growth of individuals
Pearl (1927)	logistic population growth
Fisher and Tappitt (1928)	Weibull ageing
Kleiber (1932)	respiration scales with body weight ^{3/4}
Mayneord (1932)	cube root growth of tumours
Monod (1942)	growth of bacterial populations
Emerson (1950)	cube root growth of bacterial colonies
Huggett and Widdas (1951)	foetal growth
Weibull (1951)	survival probability for ageing
Best (1955)	diffusion limitation of uptake
Smith (1957)	embryonic respiration
Leudeking and Piret (1959)	microbial product formation
Holling (1959)	hyperbolic functional response
Marr et al. (1963); Pirt (1965)	maintenance in yields of biomass
Droop (1973)	reserve (cell quota) dynamics
Rahn and Ar (1974)	water loss in bird eggs
Hungate (1975)	digestion
Beer and Anderson (1997)	development of salmonid embryos

the standard DEB model is that 26 popular empirical models for various aspects of metabolism turn out to be special cases of DEB models or very good numerical approximations, see Table 3. This suggests that the set of stylised facts constrain simple models more than is apparent on first sight, even without a formal context. Another aspect is that organisms are shaped by evolution. The ones with a simpler metabolic organisation can adapt more easily to changing environments, which can be captured by simple models. Anyway, none of these modelers realised that they had been modeling different aspects of the same thing. It is, indeed, not self evident that the development of a bird egg (Rahn and Ar, 1974) has intimate relationships with the dynamics of (nutrient) cell quota of algae (Droop, 1973) and both follow from the same set of assumptions. Nor is it clear why the method of indirect calorimetry (Lavoisier and Laplace, 1780) holds the explanation for Kleiber's law (Kleiber, 1932). DEB theory shows the crucial role of reserve in both relationships (Kooijman, 2010, Sec 4.8.2); an explanation that is hard to find if one would try to relate them directly. Unlike the method of indirect calorimetry, most ecological literature ignores the contributions of carbon dioxide and nitrogen-waste in the relationship between heat and dioxygen consumption, let alone the need to calibrate the coefficients because of substantial variation (Walsberg and Hoffman, 2005; 2006). This sloppiness of dealing with respiration probably contributed to why most of the ecological literature still struggles to understand Kleiber's law.

Another remarkable observation is that the only way yet available to access the entropy of living biomass, a notoriously difficult task indeed, is via the parameters of a DEB model. To get those parameters, we need to follow the individual throughout its life cycle and monitor what goes in and out. Gaps in knowledge of these dynamic budgets can partly be compensated by estimating parameters in context, i.e. using information from other (related) species. The estimation procedure presently has quite a few handles for this (Lika et al., 2020; Marques et al., 2018), as demonstrated by the Add-my-Pet collection of animal species from all corners of animal kingdom (Amp, 2020).

3. Discussion

A class of stoichiometric models in ecology follows chemical elements (Sterner and Elser, 2002) in ways that allow application of conservation of mass. Although stoichiometric constraints on metabolism are very important (Kooijman, 1995), the problem here is that atoms are coupled in molecules that are subjected to metabolic transformation. This coupling of transport of chemical elements forms the basis of the isotope dynamics module of DEB theory (Kooijman, 2010; Pecquerie et al., 2010, Section 4.7) and is essential to make the link to energetics. Inclusion of this coupling requires the delineation of pools of metabolites, as mentioned in step 2.

None of the assumptions in Table 1 directly relate to respiration, but nonetheless, they fully specify this frequently-used flux as quantifier for metabolic rate in aerobic environments, via conservation rules for chemical elements. A variety of processes, such as maintenance and feeding or growth overheads, contribute to respiration. This is in nice harmony with application of conservation rules, since they work out additively. Despite the fact that this additive nature of sub-fluxes of respiration is well-known and generally accepted, respiration is frequently modelled as an allometric function of body mass (Kleiber, 1932), which directly involves an inconsistency: sums of functions of body mass cannot add up as an allometric function of body mass. Attempts to include allometric respiration into growth models are bound to fail for this reason. See the comments of van der Meer (2018) and Sousa et al. (2009) on Hou et al. (2008) and of Sibly et al. (2015) on Sibly et al. (2013). This partly explains why modelling metabolism of individuals dynamically is rarely attempted. Another partial explanation is that metabolic rate is frequently seen as a driver of metabolism (Brown et al., 2004; Calder et al., 1984; Peters, 1983); the success of DEB theory questions this view.

Most published animal energy budget models classify as net production models, which are among the alternatives mentioned in step 5: maintenance is first subtracted from assimilation and the remainder is allocated to growth and/or reproduction. By identifying respiration with maintenance, as is typically done, these models ignore production overheads, and are, therefore, at odds with thermodynamics. By excluding production from respiration, this view opposes the previous view where respiration, interpreted as metabolic rate, would drive metabolism. The whole field is in need for a fundamental shake-up. If one would take the consequence and disregard all papers that use production models or allometric functions for respiration, little is left from the literature on energetics.

The take-home message of this paper is that the more simple the model is in terms of numbers of state variables and parameters, the more constraining consistency conditions become. This answers the quest mentioned in the introduction. It is possible to construct models of non-DEB nature that are much more complex, such as the 9-pool model with 40 parameters for the fungus *Penicillium* (Tang et al., 2017), which represents a mix between pool and biochemical models and conserves elemental frequencies by approximation. Where growth-and-division controls body size of microbes, with an uptake that is typically taken proportional to mass, we need more parameters for a corresponding model for animals, to deal with life stages, aging, reproduction and body size control.

It is possible, and frequently even necessary, as shown in many places (Kooijman, 2010), to extend the standard DEB model. These extensions should, I think, be seen as added modules to a common core structure, leading to a family of related models with clear family relationships. Some details can be modified without much harm for the setup, such as the replacement of the assumption for maternal effect (Table 1 assumption 4), but this costs at least an extra parameter and properties of such a model will be harder to analyze. Such minor modifications do not result in a really different model, in my opinion, but can be functional in particular cases.

What level of model complexity is actually required? The answer to

this key question depends on the context of the research aims, so there is not a single answer. Simplicity is attractive for many reasons, but too much simplicity gives problems with reality. If biodiversity cannot be accommodated, we also lose connection with the evolutionary context. Too much complexity, on the other hand, easily gives problems with practical applicability, since data will be lacking. Kooijman et al. (2008) discuss the relationship between available data and parameters that can be estimated in the context of the standard DEB model, showing that the estimation of all primary parameters of the standard DEB model already frequently suffers from the problem of lack of data. While data completeness level 10 specifies the energy balance empirically for different food levels, see Lika et al. (2011, Table 3), the mean completeness level in the AmP collection of 2026 animal species is only some 2.5, and the maximum one 6 on 2019/10/31. This motivated the development of context-based parameter estimation methods, although ideally all parameters should be determined accurately by data. This is presently way beyond scope. Much more complex models will have a strong tendency to be species-specific which hampers comparison with other species. They are also difficult to test and uncertainty about parameter values easily makes them useless. Although the reasons are very different, the implication is that the standard DEB model not only lacks equally simple, but also more complex alternatives, since this wide testing ability is a requirement for general models.

A good advice is, I think, to be aware of the 'Christmas tree

Appendix A. Generalised von Bertalanffy growth rate

Von Bertalanffy growth specifies change in length L as $\frac{d}{dt}L = \dot{r}_B(L_\infty - L)$, where L_∞ is ultimate length, and $\dot{r}_B$ is the von Bertalanffy growth rate, which is taken to be constant at constant food. The specific growth rate $\dot{r}$ is defined as $\dot{r} = \frac{d}{dt} \ln L^3$, so $\frac{d}{dt} \ln L = \dot{r}/3$, which results in the relationship $\dot{r} = 3\dot{r}_B(L_\infty/L - 1)$. The biological interpretation of $\dot{r}_B$ is that $\frac{3}{2}\dot{r}_B$ equals the specific growth rate $\dot{r}$ at maximum growth (Kooijman et al., 2020).

More generally, for length as satiating function of age at constant food, the generalised von Bertalanffy growth rate can be defined as

$$\dot{r}_B \equiv L_\infty^{-1} \frac{d}{dt} L \bigg|_{L=0} = \lim_{L \downarrow 0} \frac{\dot{r}L}{3L_\infty}$$

Notice that $\lim_{L \downarrow 0} \dot{r} = \infty$. While the (normal) von Bertalanffy growth rate applies for all lengths, the generalised von Bertalanffy growth rate $\dot{r}_B$ is a limit that only applies for $L \downarrow 0$.

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syndrome' (Kooijman, 2018): Just before Christmas, many people buy a Christmas tree for indoors; beautiful, but not beautiful enough. They start hanging balls, and other stuff, to enhance beauty, but if you continue too long with this, the Christmas tree will tumble over and you have nothing. The tree stands for the basic model structure, the balls for modules to include particular details. For me, the standard DEB model represents the tree without balls, the skeleton. My suggestion would be: don't forget to remove ornaments that are not needed (i.e. functional, matched by data availability).

Declaration of Competing Interest

There are no conflicts of interest

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